ISEP

Segurança de Sistemas e Informação (SEGSI)
PL09
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Exercise 1: Security Monitoring

How do you know if you got compromised? You can't unless you have some evidence of the happening, being in the form of logs or a ransomware note. In this assignment, we will focus on the former. You are tasked with installing a Security Information and Event Monitoring (SIEM) tool. This tool would gather information from several sources and correlate information to warn.

1. Read up and understand the tool Wazuh[1]. Understand its purpose and capabilities;
2. Install the tool using the quickstart guide[2];
3. Install two different agents: one Windows and one Linux and configure them to report to the installed Wazuh[3];
4. Take this opportunity to verify that the hardening done on your machines logs all commands executed by the user. Configure the Wazuh agent to gather those logs and index them.

The tool is not static, it needs nourishment to be of value to the enterprise. With that in mind, you are tasked with creating a few standard rules. Be at the ready with the rule and decoder guides provided by the tool documentation[4,5].

5. Create or extend a rule to raise a 12-level alert if more than 10 login attempts are done in 60 seconds, both for Windows and Linux[6].
6. Create a Linux rule to detect if a user runs tcpdump on the Linux machine. Set it to level 10.
7. Create a rule to detect if a specific management user escalates their privileges to root (choose a random user to do that). Raise a level 12 alarm.
8. Verify your rules in the dashboard itself.

[1] https://wazuh.com/
[2] https://documentation.wazuh.com/current/quickstart.html
[3] https://documentation.wazuh.com/current/user-manual/agent/agent-enrollment/index.html
[4] https://documentation.wazuh.com/current/user-manual/ruleset/ruleset-xml-syntax/decoders.html
[5] https://documentation.wazuh.com/current/user-manual/ruleset/ruleset-xml-syntax/rules.html
[6] https://documentation.wazuh.com/current/user-manual/capabilities/log-data-collection/configuration.html